

Grand Quiz

Dynamics MCQ's (Lec 1 to 22)

1. The branch of mechanics which deals with the motion of object is known as
Ans: dynamics

2. The branch of applied mathematics that deals with study of motion of body without taking into consideration how the motion is caused is;
Ans: kinematics

3. Treat of the geometry of motion of bodies without taking into account their masses or the forces acting on them.
Ans: kinematics

4. The curve that is traced by a particle as the time progresses is called the
Ans: trajectory

5. The instantaneous velocity of particle is given by:
Ans: $\lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t}$

6. The instantaneous acceleration of a particle is given by:
Ans: $\lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t}$

7. $\vec{v} = ?$
Ans: $\frac{d\vec{r}}{dt}$

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8- $\vec{a} = ?$

Ans $\frac{d\vec{v}}{dt}$ or $\frac{d^2\vec{r}}{dt^2}$

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9- If $\vec{r} = x\hat{i} + y\hat{j}$, then $\vec{v} = ?$

Ans $\frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j}$

10- If $\vec{r} = x\hat{i} + y\hat{j}$, then $|\vec{v}| = ?$

Ans $\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$

11- If $\vec{r} = x\hat{i} + y\hat{j}$, then $\vec{a} = ?$

Ans $\frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j}$

12- The Tangential component of velocity is the magnitude of the velocity.

Ans magnitude

13- The normal component of velocity is always.

Ans zero

14- At any instant the particle is moving in the direction of -

to the path

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Ans tangent

15- A particle is moving with a velocity $3\hat{i} + 7\hat{j}$ the tangential component of velocity is?

Ans 5.8

16- A particle is moving with a velocity $3\hat{i} + 7\hat{j}$, normal component of velocity is?

Ans Zero "0"

17- At any time t the position of a particle moving in a plane can be specified by $(a \cos \omega t, a \sin \omega t)$ where a and ω are constant, the tangential component of velocity is

Ans $a\omega$

18- At any time t the position of a particle moving in a plane can be specified by $(a \cos \omega t, a \sin \omega t)$, where a & ω are constant the normal component of velocity is

~~zero~~ zero

Q#2

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19- The velocity has _____ along the normal to the path

Ans- no component.

20- If v is the magnitude of velocity of a particle moving along a trajectory, then the tangential component of acceleration is

Ans $\frac{dv}{dt}$

21- The normal component of acceleration of a particle moving with the velocity magnitude v is

Ans $\frac{v^2}{r}$

22- If a particle is moving with uniform speed, then the tangential component of its acceleration is

Ans zero

23- A particle is moving with speed $v = t \sin t$, the tangential component of its acceleration is?

Ans $+ \cos t + t \sin t$.

24. A particle is moving with a speed $v_0 + 5t$, the tangential component of its acceleration at $t=0$ is

Ans Zero

25. A particle is moving with a speed $v_0 + 5t$, the tangential component of its acceleration at $t = \frac{\pi}{2}$ is

Ans 1

26. A particle is moving with a speed $v_0 \cos t$, the tangential component of its acceleration is

Ans $\cos t$

27. A particle is moving with a speed $v_0 e^t + t \cos t$, the tangential component of its acceleration at $t=0$ is

Ans 2

28. The direction of radius vector is known as

Ans Radial direction

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29. The direction perpendicular to radius vector in the direction of increasing angle is called

Ans. Transverse direction.

30. The radial component of velocity is

Ans. $\dot{r}$

31. The Transverse component of velocity is

Ans. $r\dot{\theta}$

32. The radial component of acceleration is

$$\ddot{r} - r(\dot{\theta})^2$$

33. The transverse component of acceleration is

Ans. $\frac{1}{r} \frac{d}{dt} (r^2 \dot{\theta})$ or also $2\dot{r}\dot{\theta} + r\ddot{\theta}$

34. The position of a particle moving along a path at any time t is given by $\vec{r} = \sin t \cos t \hat{i} + \sin^2 t \hat{j}$

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40- If x is the distance of a particle from a fixed point at any time t , then the magnitude of velocity is

Ans $\frac{dx}{dt}$

41- If x is the distance of a particle from a fixed point at any time t , then the magnitude of acceleration is

Ans $\frac{d^2x}{dt^2}$

42- $v = ?$

Ans $u + at$

43- $x = ?$

Ans $ut + \frac{1}{2}at^2$

44- $2ax = ?$

Ans $v^2 - u^2$

45- The distance travelled by a

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46. particle moving in a straight line with uniform acceleration a in the n^{th} unit of time, where u is initial speed of the particle is

Ans $u + \frac{1}{2} a(2n-1)$

47. The distance travelled by a particle moving in a straight line with uniform acceleration a and initial speed u in the 8th second is

Ans $u + \frac{15}{2} a$

48. The distance travelled by a particle moving in a straight line with uniform acceleration a and initial speed u in 9th second is

Ans $u + \frac{17}{2} a$

49. When the acceleration is function of time only, we may write $a(t) =$

Ans $\frac{dx}{dt^2}$

50. When the acceleration is function of velocity only, we may write $a(v) =$

Ans $v \frac{dv}{dx}$ also $v \frac{dv}{dx}$

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Q1- When the acceleration is a function of distance x only, we may write $a(x)$.

Ans $v \frac{dv}{dx}$ or also $\frac{d^2x}{dt^2}$

Q2- The slope of tangent to the curve $y = e^x \sin x$ is $\frac{dy}{dx} =$

Ans $e^x (\cos x + \sin x)$

Q3- The slope of tangent to the curve $y = e^x \sin x$ at $x=0$ is

Ans 1

Q4- The velocity of a particle at a distance x is equal to the slope of the space-time curve at a point whose coordinate is x .

Ans tangent.

Q5- For a constant velocity, the slope of space-time curve is

Ans remains the same

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56. The space-time curve is a _____ for a constant velocity.

Ans. Straight line.

57. The _____ of a particle moving in a straight line is given by the slope of the velocity-time curve.

Ans. acceleration.

58. If the acceleration is constant, the velocity-time curve will be a _____ whose slope gives the acceleration.

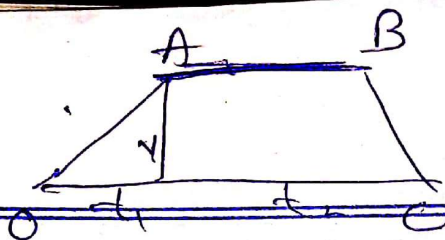
Ans. Straight line.

59. The slope of a velocity-time curve of a particle moving in a straight line gives its _____.

Ans. acceleration.

60. The area under the velocity-time curve of a particle moving in a straight line gives the _____ by the particle.

Ans. distance.



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61. OA = accelerated motion
BC = ~~retard~~ retarded motion.

$$\frac{1}{2} v (t + t_2) = \text{distance covered by particle}$$

Def of S.H.M (imp).

62. The maximum distance covered by the particle executing S.H.M in either side of mean position is called.

Ans. amplitude.

63. The time taken by the particle to complete one oscillation is called

Ans. Time Period.

64. The particle executing S.H.M has velocity at mean position

Ans. maximum.

65. The time period of a particle executing S.H.M is.

Ans. $\frac{2\pi}{\omega}$

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66. The frequency of particle executing S.H.M is

Ans $\frac{T}{2\pi}$

67. The projection of a particle moving along a circle executes.

Ans:- Simple harmonic motion

Quiz:-

68. If a force acts at a time t on a particle of mass m moving with velocity v , then the equation of motion according to Newton's second law is

Ans $\vec{F} = \frac{d}{dt}(m\vec{v})$

69. A body of mass $m = 5g$ is executing S.H.M about a point O with amplitude $10cm$. If its max velocity is $100cm/sec$. Its velocity will be $50cm/sec$ at a distance

Ans $5\sqrt{3}$

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70. A particle of mass 4 kg is in motion under the action of variable force F and is moving under velocity $3t^2$ then the force acting at time "2" second is.

Ans 48 N .

71. In a conservative field of force the total energy of particle ——— throughout the motion.

Ans remains constant.

72. A particle of mass 3 kg is moving with velocity $4\hat{i} + 5\hat{j}$. The force F acting on the particle will be.

Ans $12\hat{i}$

73. At mean position of pendulum the potential energy of pendulum is.

Ans minimum

74. If a particle of mass 4 kg is moving with speed of 3 m/s then the kinetic energy of particle is.

Ans 18 J

$$\text{Work} = F \times d$$

$$= 20 \times 3$$

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75. Force acting on a body change its kinetic energy, direction of motion, speed, all

Ans all

76. If a particle is moving along a circle of radius a with center O . Which of the following is true about transverse component of force in polar.

$$F_{\theta} = m a \ddot{\theta}$$

77. During S.H.M the kinetic energy of body at mean position is

Ans Maximum (because v is max).

78. Transverse component of force in polar coordinate.

$$F_{\theta} = m (2 \dot{r} \ddot{\theta} + r \ddot{\theta})$$

79. If x is displacement of particle from a fixed point at any time t and is given by $x = 5t^2$. Then expression for acceleration of

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of particle is

Ans

10

80- which of the following is true for a force F to be conservative

Ans

$$\text{Curl } F = 0$$

81- The rate of change of a particle about a point O is equal to the torque about O of the force on the particle

Ans

angular momentum

82- If the mass of the particle is doubled, then the gravitational force is earth's gravitational field is

Ans

doubled

83- If the velocity of the particle is $\frac{1}{2} 3t^2 + 4$ then the acceleration of particle at times "3" seconds will be

Ans

18

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84. Direction of momentum is direction of object's?

Ans. Velocity.

85. Radial component of force in polar coordinates.

Ans. $F_r = m(\ddot{r} - r(\dot{\theta})^2)$

86. Rate of work done to the particle is

Ans. Power